

Course Code- 151612**Quantum Computing****3 0 0 3****Unit-1.0:****8 hrs**

Introduction to Quantum Computing: Motivation for studying Quantum Computing, Major players in the industry (IBM, Microsoft, Rigetti, D-Wave etc.), Origin of Quantum Computing. Overview of major concepts in Quantum Computing, Qubits and multi-qubits states, Bra-ket notation, Bloch Sphere representation, Quantum Superposition, Quantum Entanglement.

Unit-2.0:**6 hrs**

Math Foundation for Quantum Computing: Matrix Algebra: basis vectors and orthogonality, inner product and Hilbert spaces, matrices and tensors, unitary operators and projectors, Dirac notation, Eigen values and Eigen vectors.

Unit-3.0:**6 hrs**

Building Blocks for Quantum Program: Architecture of a Quantum Computing platform. Details of q-bit system of information representation: Bloch Sphere, Multi-qubits States.

Unit-4.0:**6 hrs**

Quantum superposition of qubits (valid and invalid superposition), Quantum Entanglement, Useful states from quantum algorithmic perspective e.g. Bell State, Operation on qubits: Measuring and transforming using gates.

Unit-5.0:**8 hrs**

Quantum Logic gates and Circuit: Pauli, Hadamard, phase shift, controlled gates, Ising, Deutsch, swap etc. Programming model for a Quantum Computing Program: Steps performed on classical computer, Steps performed on Quantum Computer, Moving data between bits and qubits.

Unit-6.0:**8 hrs**

Quantum Algorithms: Basic techniques exploited by quantum algorithms, Amplitude amplification, Quantum Fourier Transform, Phase Kick-back, Quantum Phase estimation, Quantum Walks. Major Algorithms: Shor's Algorithm, Grover's Algorithm, Deutsch's Algorithm, Deutsch - Jozsa Algorithm. OSS Toolkits for implementing Quantum program: IBM quantum experience, Microsoft Q, Rigetti PyQuil (QPU/QVM).

Text/Reference:

1. Michael A. Nielsen, "Quantum Computation and Quantum Information", Cambridge University Press.
2. David McMahon, "Quantum Computing Explained", Wiley.
3. IBM Experience: <https://quantumexperience.ng.bluemix.net>.
4. Microsoft Quantum Development Kit <https://www.microsoft.com/en-us/quantum/development-kit>.
5. Forest SDK PyQuil: <https://pyquil.readthedocs.io/en/stable/>.